

HDOC Day-18

Question: Menu-Driven Program for Digit Operations Using User-Defined Functions

1. Problem Statement & Identification

- **Problem:** Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer using separate user-defined functions (void return type and no arguments) for each option:
 1. Check whether it is a Strong number.
 2. Check whether it is an Armstrong number.
 3. Check whether it is a Harshad number.
 4. Check whether it is a Duck number.
- **Input:** Menu choice (choice), and inside each function, a positive integer (num).
- **Output:** Display result directly inside the corresponding user-defined function.

2. Algorithm

1. **Start**
2. Display menu options (1 to 4).
3. Read user choice.

4. Execute user-defined function based on choice using switch-case:

- **Case 1: checkStrong()**

1. Read num. Set temp = num, sum = 0.
2. Extract each digit = temp % 10 and calculate its factorial (fact).
3. Add fact to sum and update temp /= 10.
4. If sum == num, print "It is a Strong number", else print "It is not a Strong number".

- **Case 2: checkArmstrong()**

1. Read num. Count total digits (count).
2. Extract each digit and compute sum += digit^count.
3. If sum == num, print "It is an Armstrong number", else print "It is not an Armstrong number".

- **Case 3: checkHarshad()**

1. Read num. Compute the sum of its digits (sum).
2. If num % sum == 0, print "It is a Harshad number", else print "It is not a Harshad number".

- **Case 4: checkDuck()**

1. Read num_str as a string.
 2. Check if num_str[0] != '0' and if it contains at least one '0' at subsequent positions.
 3. If valid, print "It is a Duck number", else print "It is not a Duck number".
- **Default:** Print "Invalid Choice!".

5. **End**

3. Test Case Table

Tes t Ca se	Cho ice	Input Num ber	Opera tion	Expec ted Outpu t	Actua l Outpu t	Stat us
1	1	145	Check Strong Numb er	It is a Strong numb er	It is a Strong numb er	Pas s
2	2	153	Check Armstr ong Numb er	It is an Armstr ong numb er	It is an Armstr ong numb er	Pas s

Tes t Ca se	Cho ice	Input Num ber	Opera tion	Expec ted Outpu t	Actua l Outpu t	Stat us
3	3	18	Check Harsh ad Numb er	It is a Harsh ad numb er	It is a Harsh ad numb er	Pas s
4	4	3205	Check Duck Numb er	It is a Duck numb er	It is a Duck numb er	Pas s

4. C Source Code

```
#include <stdio.h>
#include <math.h>

// Function Prototypes (void return type, no parameters)
void checkStrong(void);
void checkArmstrong(void);
void checkHarshad(void);
void checkDuck(void);

int main(void) {
    int choice;

    printf("--- MENU ---\n");
    printf("1. Check whether it is a Strong number\n");
    printf("2. Check whether it is an Armstrong number\n");
```

```
printf("3. Check whether it is a Harshad number\n");
printf("4. Check whether it is a Duck number\n");
printf("Enter your choice (1-4): ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        checkStrong();
        break;
    case 2:
        checkArmstrong();
        break;
    case 3:
        checkHarshad();
        break;
    case 4:
        checkDuck();
        break;
    default:
        printf("\nInvalid Choice!\n");
        break;
}

return 0;
}

void checkStrong(void) {
    int num, temp, digit, sum = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        digit = temp % 10;

        int fact = 1;
        for (int i = 1; i <= digit; i++) {
            fact *= i;
        }
    }
}
```

```

        sum += fact;
        temp /= 10;
    }

    if (sum == num) {
        printf("\nNumber: %d, Output: It is a Strong number\n", num);
    } else {
        printf("\nNumber: %d, Output: It is not a Strong number\n", num);
    }
}

void checkArmstrong(void) {
    int num, temp, digit, count = 0;
    double sum = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        count++;
        temp /= 10;
    }

    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        sum += pow(digit, count);
        temp /= 10;
    }

    if ((int)sum == num) {
        printf("\nNumber: %d, Output: It is an Armstrong number\n", num);
    } else {
        printf("\nNumber: %d, Output: It is not an Armstrong number\n",
num);
    }
}

void checkHarshad(void) {

```

```

int num, temp, digit, sum = 0;

printf("Enter a positive integer: ");
scanf("%d", &num);

temp = num;
while (temp > 0) {
    digit = temp % 10;
    sum += digit;
    temp /= 10;
}

if (sum != 0 && num % sum == 0) {
    printf("\nNumber: %d, Output: It is a Harshad number\n", num);
} else {
    printf("\nNumber: %d, Output: It is not a Harshad number\n", num);
}
}

void checkDuck(void) {
    char num_str[100];
    int is_duck = 0;

    printf("Enter a positive integer: ");
    scanf("%s", num_str);

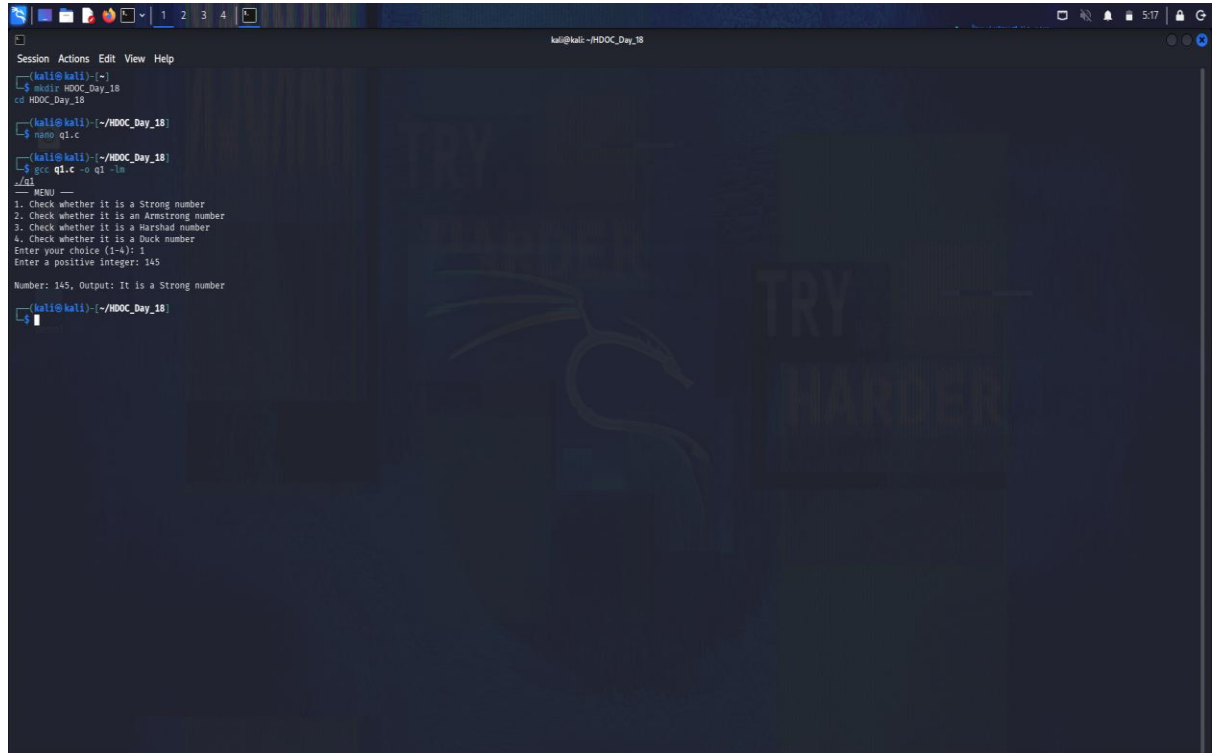
    if (num_str[0] != '0') {
        for (int i = 1; num_str[i] != '\0'; i++) {
            if (num_str[i] == '0') {
                is_duck = 1;
                break;
            }
        }
    }

    if (is_duck) {
        printf("\nNumber: %s, Output: It is a Duck number\n", num_str);
    } else {
        printf("\nNumber: %s, Output: It is not a Duck number\n", num_str);
    }
}

```

}

6. Compilation, Execution & Screenshots



```
kali@kali: ~/HDOOC_Day_18
$ mkdir HDOOC_Day_18
$ cd HDOOC_Day_18
$ nano q1.c
$ gcc q1.c -o q1 -lm
./q1
-- MENU --
1. Check whether it is a Strong number
2. Check whether it is an Armstrong number
3. Check whether it is a Harshad number
4. Check whether it is a Duck number
Enter your choice (1-4): 1
Enter a positive integer: 145
Number: 145, Output: It is a Strong number
$
```